LEARNER GUIDE

The perpendicular bisector

Constructions, Loci and Bearings · Lesson 3. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Imagine two friends at points A and B who want to meet somewhere the **same distance** from both homes. There is not one fair spot but a whole **line** of them — and every point on that line is exactly the same distance from A as from B. That line is the **perpendicular bisector** of AB. In this lesson you discover it by hunting for equal-distance points, then construct it exactly with equal compass arcs, and use it to find a midpoint or a circle through two points.

What you are learning to notice

- A point the same distance from A and B is **equidistant** from them.
- **All** the equidistant points together form a straight line — the **perpendicular bisector** of AB.
- It is **perpendicular** to AB (a right angle) and passes through the **midpoint** of AB.
- Equal arcs from A and B cross at points equidistant from both, so the line through those crossings is the bisector.

Moving through it, one step at a time

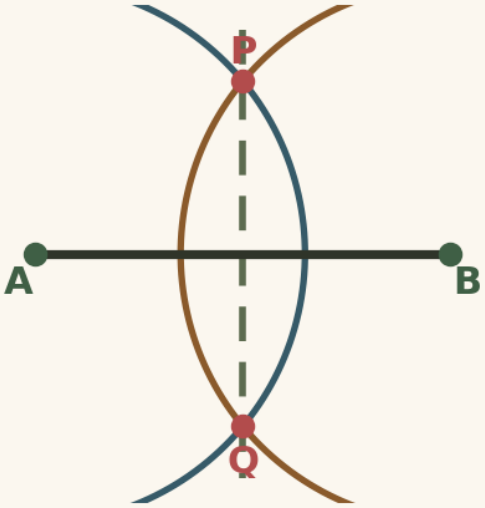
To construct it: open the compass wider than half of AB. From A, swing an arc above and below the line; keep the same opening and swing an arc from B. The arcs cross at two points — each is the same distance from A and from B. Draw the straight line through those two crossings: that is the perpendicular bisector. It meets AB at the midpoint. Leave your arcs showing.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Construct the perpendicular bisector of AB

Construct the perpendicular bisector of the segment AB.

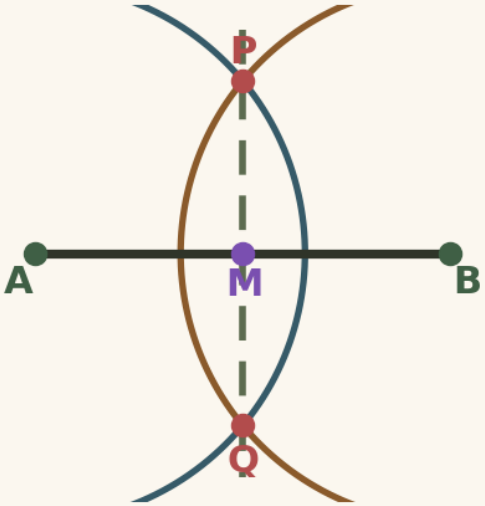


1. Open the compass wider than half of AB.	radius > ½ AB
2. Swing an equal arc from A (above and below).	arc from A
3. Keep the same opening; swing an arc from B.	arc from B
4. Mark the two crossings P and Q; draw the line PQ.	join P–Q

PQ is the perpendicular bisector: every point on it is the same distance from A and from B.

Example 2 — Use it to find the midpoint

The perpendicular bisector meets AB at its exact midpoint.



1. Construct the perpendicular bisector of AB (as above).	line PQ
2. Mark M where PQ crosses AB.	M on AB
3. M is the midpoint: MA = MB.	MA = MB

M is the exact midpoint of AB, found without measuring — MA = MB.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (equal-distance compass arcs (Lesson 1), and the distance between two points), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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